

RMO(INDIA) – 1998

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABCD be a convex quadrilateral in which $m\angle BAC = 50^\circ$, $m\angle CAD = 60^\circ$, $m\angle CBD = 30^\circ$ and $m\angle BDC = 25^\circ$. If E is the point of intersection of AC and BD, find $m\angle AEB$.
 2. Let n be a positive integers and $p_1, p_2, \dots, p_n$ be n prime numbers all larger than 5 such that 6 divides $p_1^2 + p_2^2 + \dots + p_n^2$. Prove that 6 divides n.
 3. Prove that for every natural number $n > 1$ the following inequality:

$$\frac{1}{n+1} \left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1} \right) > \frac{1}{n} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right)$$
 4. Let ABC be a triangle with $AB = AC$ and $m\angle BAC = 30^\circ$. Let A' be the reflection of A in the line BC; B' be the reflection of B in the line AC; C' be the reflection of C in the line AB. Show that A' , B' , C' form the vertices of an equilateral triangle.
 5. Find the minimum possible least common multiple (lcm) of twenty (not necessarily distinct) natural numbers whose sum is 801.
 6. Given the 7-elements set $A = \{a, b, c, d, e, f, g\}$, find a collection T of 3-element subset of A such that each pair of elements from A occurs exactly in one of the subset of T.

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